

2.1 The Periodic Table: Chemical Periodicity

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.1 The Periodic Table: Chemical Periodicity
Difficulty	Medium

Time allowed: 70

Score: /57

Percentage: /100

The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements in the Periodic Table.

[illegible]

i) Using Fig. 1.1 identify the element that forms a soluble hydroxide and an insoluble sulfate.

[1]

ii)
Using Fig. 1.1 identify the most volatile element in a group that contains elements in all three states of matter at room temperature and pressure.

[7]

iii)
Using Fig. 1.1 identify the element that forms the largest cation.

[7]

[3 marks]

Question 1b

Fig. 1.2 shows the relative first ionisation energies of six successive elements in the Periodic Table.

The letters are **not** the symbols of the elements.

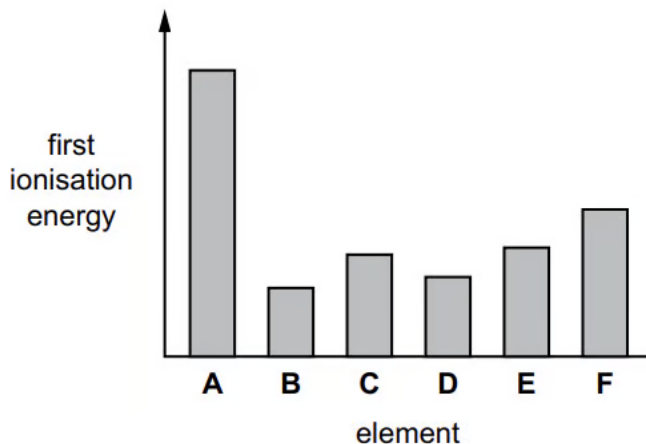


Fig. 1.2

i)
Define first ionisation energy.

[2]

ii)
Suggest why the first ionisation energy of **B** is much less than that of **A** in Fig 1.2.

[3]

[5 marks]

Question 1c

- i)
On Fig. 1.3, sketch a graph to show the trend in the atomic radius of successive elements in Period 3.

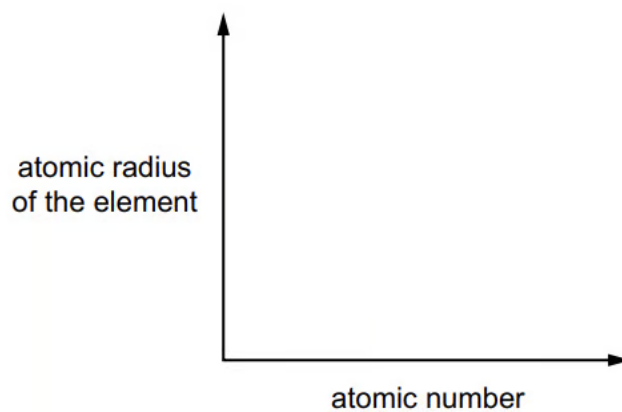


Fig. 1.3

[1]

- ii)
Explain your answer to (c)(i).

[3]

[4 marks]

Question 2a

Oxides are compounds which usually contain oxygen combined with one other element.

Oxides are classified as follows.

acidic	alkaline	amphoteric	basic
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Complete Table 2.1 to describe the oxides of the elements of the third period of the Periodic Table from sodium to sulfur.

Table 2.1

Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂	Cl ₂ O ₇

[4 marks]

Question 2b

Some of the oxides produced can react with water.

i)

Write an equation for the reaction of each of these oxides with water.

Na₂O

SO₂

[2]

ii)

What is the pH of the resulting solution in each case?

Na₂O

SO₂

[1]

[3 marks]

Question 2c

Give the oxidation number for the Period 3 element in each oxide.

Na₂O

SO₂

[2 marks]

Question 2d

The melting points of the two Period 3 oxides are displayed in Table 2.2.

Table 2.2

Na ₂ O	1132 °C
SO ₂	-72 °C

Explain the difference in melting points.

You should refer to the structure and bonding of each substance.

[4 marks]

Question 3a

The elements in Period 3 of the Periodic Table show variations in their behaviour across the period.

Fig. 3.1 shows the variation of melting points of the elements across Period 3.

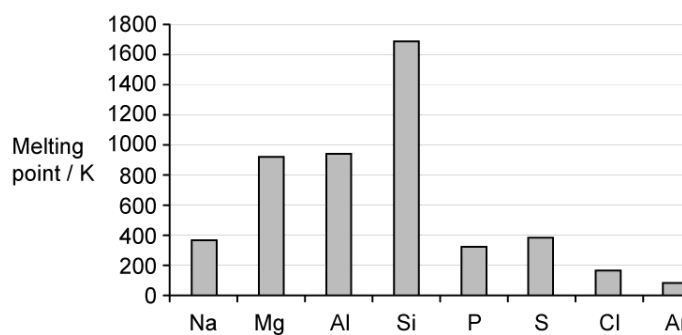


Fig. 3.1

In each of the following parts of this question you should clearly identify the interactions involved and, where appropriate, explain their relative magnitudes.

Explain the general increase in melting point from Na to Al.

[3 marks]

Question 3b

Explain the variation of melting points from P to Ar.

[3 marks]

Question 3c

Explain why Si has a much higher melting point than any of the other elements in the period.

[1 mark]

Question 3d

Going across Period 3, a trend in electronegativity can also be observed.

i)

Define the term electronegativity.

[2]

ii)

Complete Fig. 3.2 to show how electronegativity changes along Period 3.

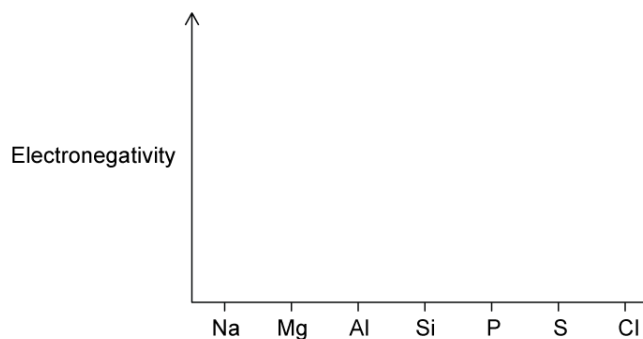


Fig. 3.2

[1]

iii)

Explain your answer to part (ii).

[2]

[5 marks]

Question 4a

This question concerns the chlorides of the elements sodium to phosphorus of the third period of the Periodic Table.

The melting points of these chlorides are given below in Table 4.1

Table 4.1

compound	Sodium chloride	Magnesium chloride	Aluminium chloride	Silicon tetrachloride	Phosphorus(V) chloride
melting point / K	1081	987	451*	203	435

*sublimes at 451 K

Give the equation, with state symbols, for the reaction of phosphorus with chlorine to form phosphorus(V) chloride, PCl_5 .

[2 marks]

Question 4b

Suggest, in terms of the structure and bonding, explanations for the following.

You should draw diagrams where you think they will help your answer.

i)

The high melting point of sodium chloride

[2]

ii)

The low melting point of silicon tetrachloride

[2]

[4 marks]

Question 4c

Write an equation for the reaction of silicon tetrachloride with water.

[1 mark]**Question 4d**

What is the pH of the solution formed when each of the following compounds is dissolved in water?

NaCl

PCl₅

[2 marks]

Question 5a

Elements in the same period of the Periodic Table show trends in physical and chemical properties.

The ionic radius for the Period 3 elements are in Table 5.1

Table 5.1

	Na ⁺	Mg ²⁺	Al ³⁺	Si ⁴⁺	P ³⁻	S ²⁻	Cl ⁻
Ionic radius (nm)	0.095	0.065	0.050	0.041	0.212	0.184	0.181

Complete Fig 5.1 to show the variation in ionic radius of Period 3 elements.

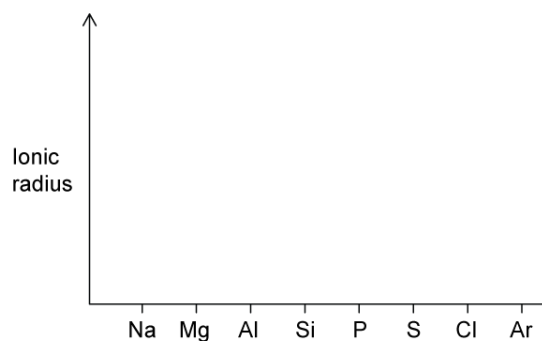


Fig. 5.1

[1 mark]

Question 5b

i)

Complete Fig. 5.2 to show the variation of electrical conductivity across Period 3.

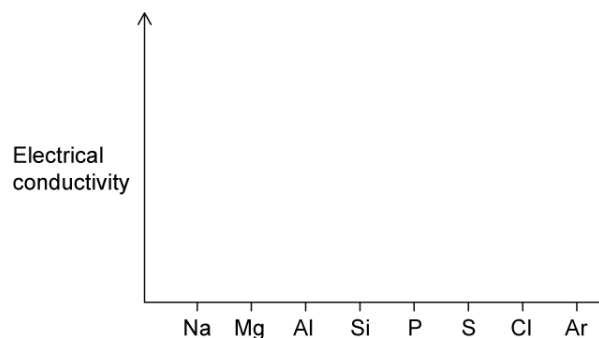


Fig. 5.2

[2]

ii)

Explain your answer to part (i)

[3]

[5 marks]

Question 5c

Period 3 elements can react with oxygen to form oxides.

Some Period 3 elements can react with water.

In the boxes below write the formulae of one of the oxides of each of these five elements.

sodium	magnesium	aluminium	phosphorus	sulfur

[2 marks]**Question 5d**

Explain why aluminium oxide is known as amphoteric.

You should use equations in your answer.

[3 marks]

Model Answers: Medium

1a

a)

i) The element that forms a soluble hydroxide and an insoluble sulfate is:

- Ba; [1 mark]

ii) The most volatile element in a group that contains elements in all three states of matter at room temperature and pressure is:

- F; [1 mark]

iii) The element that forms the largest cation is:

- Cs; [1 mark]

[Total: 3 marks]

- You should be aware of the trend in solubility in Group 2 hydroxides and sulfates:
 - Going down Group 2, the solubility of the hydroxides decreases
 - Going down Group 2, the solubility of the sulfates increases
- Group 17 is the only group which contains elements in all three states of matter at room temperature and pressure:
 - Fluorine = gas
 - Chlorine = gas
 - Bromine = liquid
 - Iodine = solid
 - Astatine = solid
 - Volatility refers to how easily a substance can evaporate
 - The lower the melting and boiling point, the more volatile a substance is
 - Fluorine is a gas and so has the lowest melting / boiling point in the group
- Metals form cations (positively charged ions) and non-metals form anions (negatively charged ions)
 - For part iii) this eliminates the elements on the right hand side of the Periodic Table

- Caesium and barium will form the largest cations as they have more shells than the elements further up the groups
- They are both in the same period:
 - Going across a period, the nuclear charge increases, so barium has a larger nuclear charge than caesium enabling it to attract the outer electrons to itself more than caesium and resulting in a smaller ion

1b

b)

i) The first ionisation energy is:

Award 1 mark for 2 or 3 points.

Award 2 marks for 4 points.

- Energy required / energy change
- When one electron is removed
- From each atom in one mole of
- Gaseous atoms

ii) The first ionisation energy of **B** is much less than that of **A** in Fig 1.2 because:

- (For element **B**) the outer electron is removed from a higher energy level; [1 mark]
- There is more shielding; [1 mark]
- So less attraction to nucleus; [1 mark]

[Total: 5 marks]

- The four factors which affect the first ionisation energy are:
 - Nuclear charge
 - Distance of outer electrons from nucleus
 - Shielding by inner electrons
 - Spin-pair repulsion
- The question tells you that the ionisation energies are for **six successive elements**
 - Generally, across a period, the ionisation energy will increase due to an increased nuclear charge and reasonably constant shielding but you must be able to explain dips like the case with element **B**
 - The decrease in ionisation energy from **A** to **B** indicates that the outer electron

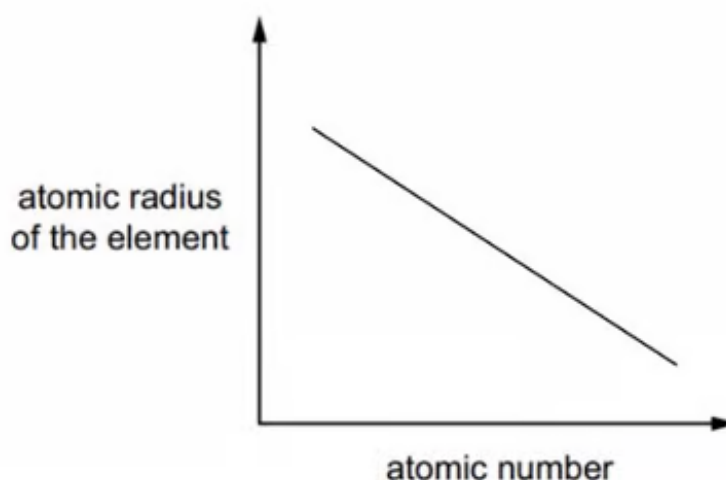
in **B** is removed from an energy level further away from the nucleus

- o This, and the fact there will be more shielding by inner electrons means the attraction between the outer electron and the nucleus is weaker and the electron is more easily lost

1c

c)

i) A graph to show the trend in the atomic radius of successive elements in Period 3 is:



- Line on graph decreases from left to right; [1 mark]

ii) To explain the trend:

- Increasing nuclear charge; [1 mark]
- Electrons in same shell; [1 mark]
- Greater attraction between nucleus and electrons; [1 mark]

[Total: 4 marks]

- Atomic radius is the distance between the nucleus and outermost electron
- Going across Period 3, there are more protons in the nucleus for each element which means a greater nuclear charge but the number of electron shells remains the same at 3
- The nucleus in each successive element in Period 3, can pull the electrons closer to

them due to the increased nuclear attraction between the outer electron and the nucleus, reducing the atomic radius

2a

a) The completed table is:

Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂	Cl ₂ O ₇
basic	basic	amphoteric	acidic	acidic	acidic	acidic

- Na₂O is basic; [1 mark]
- MgO is basic; [1 mark]
- Al₂O₃ is amphoteric; [1 mark]
- SiO₂ P₄O₁₀ and SO₂ are acidic; [1 mark]

[Total: 4 marks]

- Make sure you learn this trend
- Amphoteric is the ability to act as both an acid and a base

2b

b)

i) An equation for the reaction of each of these oxides with water is:

- Na₂O + H₂O → 2NaOH; [1 mark]
- SO₂ + H₂O → H₂SO₃; [1 mark]

ii) The pH of the resulting solution in each case is:

- Na₂O: 12 to 14
AND
SO₂: 1 to 4; [1 mark]

[Total: 3 marks]

- Sodium oxide and water react to give a highly alkaline hydroxide solution so the pH must be above 12
- Similarly, sulfur dioxide and water react to give a highly acidic solution so the pH must be very low, 1–4

2c

c) The oxidation number for the Period 3 elements in each oxide are:

- (Na) +1; [1 mark]
- (S) +4; [1 mark]

[Total: 2 marks]

- The oxidation number of Period 3 elements in an oxide corresponds to their group number
- For Period 3 oxides, the sum of the oxidation numbers of the elements will equal 0
- In an oxide, oxygen always has an oxidation number of -2
- In Na_2O , if O has an oxidation number of -2, each Na has an oxidation number of +1 to equal 0 overall
- In SO_2 , the oxidation number of oxygen is -2
 - There are two oxygen atoms so -4
 - There is only one sulfur atom which must be +4 to equal 0

2d

d) Compare the melting points of bonding and structure of Na_2O and SO_2 :

- Na_2O has a giant ionic lattice; [1 mark]
- Na_2O has strong electrostatic attraction between positive and negative ions

OR

Lots of energy is needed to break the strong attraction between oppositely charged ions; [1 mark]

- SO_2 has a simple molecular structure; [1 mark]
- SO_2 has weak instantaneous dipole-induced dipole forces

OR

It is easy to overcome the instantaneous dipole-induced dipole forces ; [1 mark]

[Total: 4 marks]

- It is logical to deduce that a four mark question asking about two compounds

- Na_2O has a giant ionic lattice; [1 mark]
 - Na_2O has strong electrostatic attraction between positive and negative ions
- OR**

Lots of energy is needed to break the strong attraction between oppositely charged ions; [1 mark]

- SO_2 has a simple molecular structure; [1 mark]
 - SO_2 has weak instantaneous dipole-induced dipole forces
- OR**

It is easy to overcome the instantaneous dipole-induced dipole forces ; [1 mark]

[Total: 4 marks]

- It is logical to deduce that a four mark question asking about two compounds needs two statements about each one to achieve full marks
- It is a good idea to learn the different structures each of the Period 3 oxides have as you go across the periodic table as you are often asked to compare them and could be in the form of a 6 mark question

3a

a) The general increase in melting point from Na to Al is because:

- The elements have metallic bonding; [1 mark]
- The strength of the attraction between the positive metal ions and delocalised electrons increases

OR

More energy is needed to break the attraction between the positive metal ions and delocalised electrons; [1 mark]

- Due to an increasing number of delocalised electrons / increasing cation charge / charge density; [1 mark]

[Total: 3 marks]

- Think about the group of the Periodic Table each element is in
 - Sodium is in Group 1 so each atom donates one electrons into the 'sea' of delocalised electrons forming Na^+ ions
 - Magnesium is in Group 2 so each atom donates two forming Mg^{2+} ions
 - Aluminium is in Group 3 so each atom donates three, forming Al^{3+} ions
 - The larger charge on the aluminium ions allows for a greater electrostatic attraction between the nucleus and electrons, needing more energy to

overcome it and consequently a higher melting point

3b

b) The variation of melting points from P to Ar is because:

- All contain instantaneous dipole-induced dipole forces between molecules; [1 mark]

- These forces are greatest in sulfur

OR

The relative magnitude of these forces is $S > P > Cl > Ar$; [1 mark]

- (Because) sulfur has the greatest number of electrons

OR

The number of electrons (in the molecules) decreases; [1 mark]

[Total: 3 marks]

- Although S, P and Cl have strong covalent bonds between the atoms, it is the van der Waals' forces between the molecules (in this case, instantaneous dipole-induced dipole forces) that are overcome at melting point
- The stronger the instantaneous dipole-induced dipole forces between molecules the more energy needed to overcome them and the higher the melting point
 - The more electrons a molecule has, the stronger the instantaneous dipole-induced dipole forces
- Consider the size of the molecules going from P to Ar when answering this question:
 - P_4 , S_8 , Cl_2 , Ar
 - Sulfur molecules are the largest with the most electrons and therefore stronger instantaneous dipole-induced dipole forces between molecules
 - Argon exists as single atoms so the instantaneous dipole-induced dipole forces will be very weak in comparison

3c

c) Si has a much higher melting point than any of the other elements in the period because:

- Si has a giant covalent structure

OR

There are strong covalent bonds; [1 mark]

[Total: 1 mark]

- Silicon has strong covalent bonds between the atoms
- Covalent bonds are much stronger than any van der Waals' force needing more energy to overcome them
- It is important you learn the trends across Period 3, they could be asked in a similar manner to this question, broken down into steps, or in the form of a 6 mark question

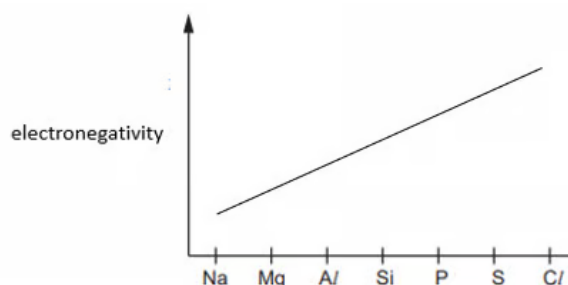
3d

d)

i) Electronegativity is defined as:

- The ability of an atom to attract electrons / a pair of electrons to itself; [1 mark]
- In a covalent bond; [1 mark]

ii) A sketch to show how electronegativity changes along Period 3:



- Line increasing; [1 mark]

iii) Explaining your answer to part (ii):

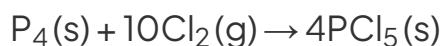
- (Going across Period 3) nuclear charge increases
AND
Atomic radius decreases; [1 mark]
- (But) there is similar shielding; [1 mark]

[Total: 5 marks]

- You can deduce from part (i) being one mark that the trend must be a simple one i.e either a single line increasing or decreasing
- The nuclear charge refers to the charge in the nucleus of the atom
- Going across a period / row in the Periodic Table, each element has one more proton than the one before it increasing the charge in the nucleus but the number of energy levels remains the same
- If the nuclear charge and atomic radius decreases, but shielding by inner electrons remains the same across the Periodic Table, electrons will be more strongly attracted to the nucleus of the atom
- Notice the question doesn't identify the trend so this should be your first marking point

4a

a) The equation, with state symbols, for the reaction of phosphorus with chlorine to form phosphorus(V) chloride, PCl_5 is:



OR



- Correct equation; [1 mark]
- State symbols; [1 mark]

[Total: 2 marks]

- You should be aware of the state of phosphorus and chlorine
- If you are unsure for phosphorus chloride, use the melting point data in the table to help you
 - The melting point is 435 K or 162 °C so must be a solid at room temperature

4b

b)

i) Sodium chloride has a high melting point because:

- It has a giant ionic lattice; [1 mark]
- Strong ionic bonds; [1 mark]

ii) Silicon tetrachloride has a low melting point because:

- It has a simple molecular structure; [1 mark]
- With weak instantaneous dipole-induced dipole forces; [1 mark]

[Total: 4 marks]

- You should be familiar with some of this content from the GCSE course
- The only difference is being more specific about the type of intermolecular force (also known as van der Waals' forces) that are being overcome in silicon tetrachloride
- **Remember:** Substances with **HIGH** melting points have **GIANT** structures, and those with **LOW** melting points have **SIMPLE** structures

4c

c) An equation for the reaction of silicon tetrachloride with water is:

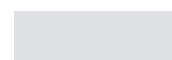
- $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$
OR
 $\text{SiCl}_4 + 4\text{H}_2\text{O} \rightarrow \text{Si}(\text{OH})_4 + 4\text{HCl}$
OR
 $\text{SiCl}_4 + 4\text{H}_2\text{O} \rightarrow \text{SiO}_2 \cdot 2\text{H}_2\text{O} + 4\text{HCl}$; [1 mark]

[Total: 1 mark]

- The standard answer for this question is the first equation
- Silicon chloride is hydrolysed in water to form a white precipitate of SiO_2 and white fumes of hydrogen chloride
- You must be able to write equations for the reaction of any of the Period 3 chlorides and water

4d

d) The pH of the solution formed when each of the following compounds is dissolved in water is:



[Total: 4 marks]

- You should be familiar with some of this content from the GCSE course
- The only difference is being more specific about the type of intermolecular force (also known as van der Waals' forces) that are being overcome in silicon tetrachloride
- **Remember:** Substances with **HIGH** melting points have **GIANT** structures, and those with **LOW** melting points have **SIMPLE** structures

4c

c) An equation for the reaction of silicon tetrachloride with water is:

- $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$
OR
 $\text{SiCl}_4 + 4\text{H}_2\text{O} \rightarrow \text{Si(OH)}_4 + 4\text{HCl}$
OR
 $\text{SiCl}_4 + 4\text{H}_2\text{O} \rightarrow \text{SiO}_2 \cdot 2\text{H}_2\text{O} + 4\text{HCl}$; [1 mark]

[Total: 1 mark]

- The standard answer for this question is the first equation
- Silicon chloride is hydrolysed in water to form a white precipitate of SiO_2 and white fumes of hydrogen chloride
- You must be able to write equations for the reaction of any of the Period 3 chlorides and water

4d

d) The pH of the solution formed when each of the following compounds is dissolved in water is:

- NaCl = pH 7; [1 mark]
- PCl_5 = pH between 1 and 4; [1 mark]

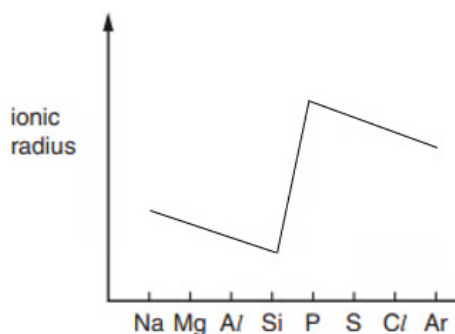
[Total: 2 marks]

- You will score the mark for sodium chloride if you state the solution is neutral
- However, you will not score the mark for saying the solution formed by PCl_5 is acidic because the solution formed is **strongly** acidic and the pH given needs to

reflect that

5a

a) The variation of ionic radius across Period 3 is:



- Decreasing from Na^+ to Si^{4+} ; [1 mark]
- Sharp increase at P^{3-} ; [1 mark]
- Decrease from P^{3-} to Cl^- ; [1 mark]

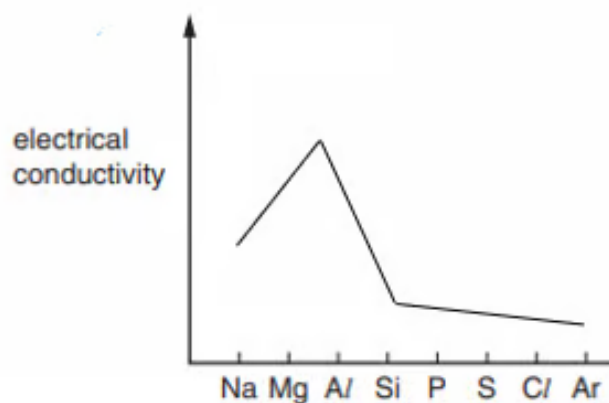
[Total: 3 marks]

- Ionic radius refers to the distance between the outermost electron of an ion
 - Na^+ to Si^{4+} are cations
 - Ionic radius increases from Na^+ to Si^{4+} because the nuclear charge increases but shielding remains the same so the electrons can be pulled more tightly towards the nucleus reducing the radius
 - P^{3-} to Cl^- are anions
 - They have gained electrons so will naturally have a larger radius than the cations
 - The decrease in ionic radius from P^{3-} to Cl^- is due to less electrons being gained but the nuclear charge still increasing

5b

b)

i) The variation of electrical conductivity across Period 3 is:



- Line increasing from Na-Al; [1 mark]
- P-Ar decreasing and (below Na); [1 mark]

ii) Explaining answer to part (i)

- Na, Mg and Al have increasing number of outer shell electrons; [1 mark]
- Si is a metalloid / semi-metal/ semi conductor; [1 mark]
- P, S and Cl are covalent / simple molecular; [1 mark]

[Total: 5 marks]

- You will not be given data in this question like you did for ionic radius
 - The variation in electrical conductivity is one you should know when you apply your understanding of bonding and structure
- For a substance to conduct electricity it needs to have a flow of charge, usually electrons
- Sodium, magnesium and aluminium all consist of positive metal ions surrounding by a sea of delocalised electrons
 - Aluminium is in Group 3 so has three outer electrons to contribute to the 'sea' compared to magnesium which has two, and sodium which has one
 - The more outer electrons in a metal, the better the conductivity of the metal due to the increased flow of charge
- Silicon has a giant molecular structure with no delocalised electrons that can move
 - It is known as a metalloid or semi conductor as it has some properties similar to metals, and others to non-metals
 - This term does not feature on the specification but has cropped up in mark schemes from previous years so make sure you learn it for the context of this question

- Simple molecular structures do not have any delocalised electrons, so do not conduct either
 - You can either draw a slightly sloping line from Si to Ar or a straight line and will achieve the mark either way

5c

c) The formula for the oxides are:

sodium	magnesium	aluminium	phosphorus	sulfur
Na₂O	MgO	Al₂O₃	P₂O₅	SO₂

- All formulae correct; [1 mark]

[Total: 1 mark]

- Alternative answers for phosphorus are P₄O₁₀ or P₂O₃
- An alternative answer for sulfur is SO₃

5d

d) Aluminium oxide is amphoteric because:

- It reacts with both acids and bases; [1 mark]
- $\text{Al}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$; [1 mark]
- $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$; [1 mark]

[Total: 3 marks]

- You could also achieve the mark if you wrote an equation for the reaction between aluminium oxide and hydrochloric acid:
 - $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + \text{H}_2\text{O}$
- The equation for the reaction between aluminium oxide and sodium hydroxide is the only one you are required to know as part of the specification